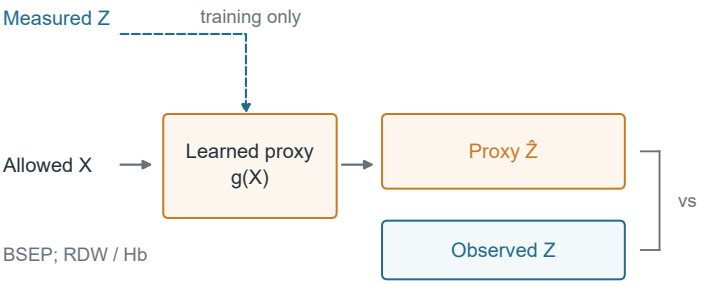
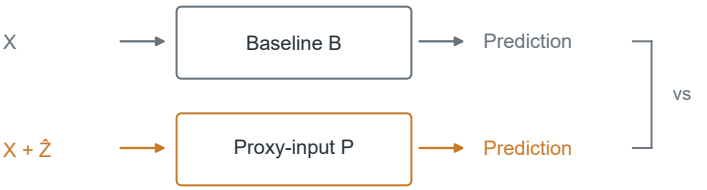
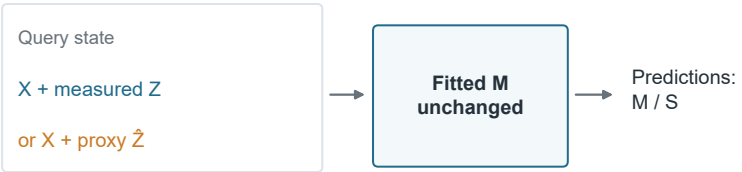
Liver-safety applications

Clinical measurements

QUESTION	COMPARISON	EVIDENCE
Q1 **Predict the measurement**	 BSEP; RDW / Hb	**Prediction error / discrimination** Proxy versus observed value
Q2 **Improve downstream prediction**	**Two separately fitted models**  DILI concern; NHANES mortality	**Increment over baseline** Same cases and outcome Y
Q3 **Evaluate input replacement**	**First establish measured-input use and benefit** Two queries of the same fitted model  PHH–rat ALT / constructed control; NHANES	**Measured gain** $G = L_B - L_M$ **Substitution loss** $D = L_S - L_M$ **Remaining gain** $H = L_B - L_S$

Establish measured-input use and benefit before interpreting replacement.